

Linear Regression Questions and Answers

Simple Linear Regression (Q1–Q7)

Q1. What is Simple Linear Regression?

Answer:

Simple Linear Regression is a statistical method used to model the relationship between one independent variable (X) and one dependent variable (Y).

$$Y = mX + c$$

Example: Predicting salary based on years of experience.

Q2. What are the key assumptions of Simple Linear Regression?

Assumption	Meaning
Linearity	Relationship between X and Y is linear
Independence	Observations are independent
Homoscedasticity	Constant variance of residuals
Normality	Residuals are normally distributed

Q3. What does the coefficient m represent in the equation $Y = mX + c$?

Answer:

The coefficient m is called the slope.

It represents the change in Y for a one-unit change in X .

$$m = \frac{\Delta Y}{\Delta X}$$

Q4. What does the intercept c represent in the equation $Y = mX + c$?

Answer:

The intercept c represents the value of Y when $X = 0$.

Q5. How do we calculate the slope m in Simple Linear Regression?

Answer:

$$m = \frac{\sum((X - \bar{X})(Y - \bar{Y}))}{\sum((X - \bar{X})^2)}$$

Where:

$\bar{X}$ = Mean of X

$\bar{Y}$ = Mean of Y

Q6. What is the purpose of the least squares method?

Answer:

The least squares method finds the best-fit line by minimizing the sum of squared residuals.

$$\sum (Y_{actual} - Y_{predicted})^2$$

Q7. How is R^2 interpreted in Simple Linear Regression?

Answer:

R^2 measures how much variation in Y is explained by X.

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

Multiple Linear Regression (Q8–Q22)

Q8. What is Multiple Linear Regression?

Answer:

Multiple Linear Regression models the relationship between multiple independent variables and one dependent variable.

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n + \epsilon$$

Q9. Difference between Simple and Multiple Linear Regression

Feature	Simple Linear	Multiple Linear
Variables	1	2 or more
Formula	$Y = mX + c$	$Y = \beta_0 + \beta_1 X_1 + \dots$

Q10. Key assumptions of Multiple Linear Regression

Answer:

1. Linearity
2. Independence
3. Homoscedasticity
4. Normality
5. No multicollinearity

Q11. What is heteroscedasticity?

Answer:

Heteroscedasticity occurs when residual variance is not constant.

Effects include:

- Unreliable standard errors
- Invalid hypothesis tests
- Poor confidence intervals

Q12. How can you improve a model with high multicollinearity?

Method	Description
Remove variables	Drop correlated features
Combine variables	Create a single feature
Ridge/Lasso	Use regularization
PCA	Reduce dimensions

Q13. Common techniques for transforming categorical variables

Technique	Description
One-Hot Encoding	Binary columns
Label Encoding	Numeric labels
Ordinal Encoding	Ordered values
Target Encoding	Mean target values

Q14. Role of interaction terms in Multiple Linear Regression

Answer:

Interaction terms capture the combined effect of two variables.

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 (X_1 X_2)$$

Q15. Interpretation of intercept in Simple vs Multiple Regression

Model	Meaning
Simple Regression	Y when $X = 0$
Multiple Regression	Y when all X values = 0

Q16. What is the significance of slope in regression analysis?

Answer:

The slope represents the change in Y for a one-unit change in X.

Q17. How does intercept provide context?

Answer:

The intercept provides the baseline value of Y when all predictors are zero.

Q18. Limitations of using R^2 alone

Limitation	Explanation
Always increases	More variables increase R^2
Ignores overfitting	High R^2 may still fail on new data
Sensitive to outliers	Outliers distort results

Q19. Interpretation of large standard error

Answer:

A large standard error means the coefficient estimate is unreliable and less precise.

Q20. How to identify heteroscedasticity?

Pattern	Interpretation
Random scatter	Good
Funnel shape	Heteroscedasticity
Curved pattern	Non-linearity

Q21. High R^2 but low Adjusted R^2

Answer:

The model likely contains unnecessary variables and may be overfitting.

Q22. Importance of scaling variables

Reason	Benefit
Faster convergence	Improves optimization
Fair comparison	Coefficients become comparable
Better regularization	Required for Ridge/Lasso

Polynomial Regression (Q23–Q31)

Q23. What is Polynomial Regression?

Answer:

Polynomial Regression models non-linear relationships.

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \cdots + \beta_n X^n + \epsilon$$

Q24. Difference between Linear and Polynomial Regression

Feature	Linear	Polynomial
Shape	Straight line	Curved line
Flexibility	Low	High
Risk	Underfitting	Overfitting

Q25. When is Polynomial Regression used?

- Non-linear relationships
- Curved trends
- Growth and decay patterns
- Saturation effects

Q26. General equation for Polynomial Regression

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \cdots + \beta_n X^n + \epsilon$$

Q27. Can Polynomial Regression be applied to multiple variables?

Answer: Yes.

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1^2 + \beta_4 X_2^2 + \beta_5 (X_1 X_2)$$

Q28. Limitations of Polynomial Regression

Limitation	Explanation
Overfitting	Fits noise
Extrapolation issues	Poor future predictions
Multicollinearity	Polynomial terms correlate
Interpretability	Hard to explain

Q29. Methods to evaluate polynomial degree

- Cross-validation
- Adjusted R^2
- AIC/BIC
- Train-test split
- Learning curves

Q30. Why is visualization important?

Purpose	Benefit
Check fitting	Validate curve
Detect overfitting	Avoid noisy fits
Explain results	Better understanding

Q31. Polynomial Regression in Python

```
import numpy as np
from sklearn.preprocessing import PolynomialFeatures
from sklearn.linear_model import LinearRegression
```

```
X = np.array([1,2,3,4,5]).reshape(-1,1)
y = np.array([1,4,9,16,25])
```

```
poly = PolynomialFeatures(degree=2)
X_poly = poly.fit_transform(X)
```

```
model = LinearRegression()
model.fit(X_poly, y)
```

```
y_pred = model.predict(X_poly)
```

Quick Reference

Topic	Formula
Simple Linear Regression	$Y = mX + c$
Multiple Linear Regression	$Y = \beta_0 + \beta_1 X_1 + \dots$
Polynomial Regression	$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \dots$
R^2 Formula	$1 - \frac{SS_{res}}{SS_{tot}}$